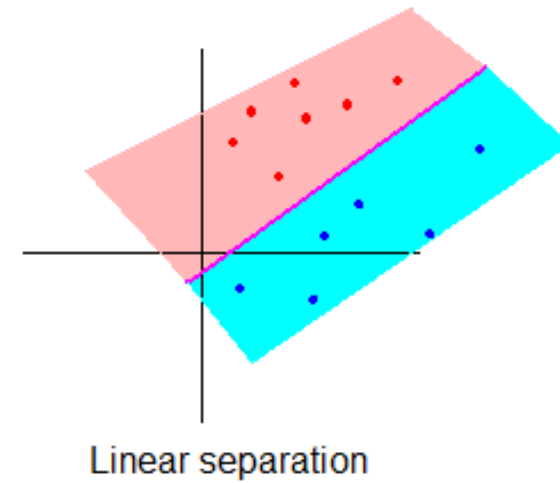
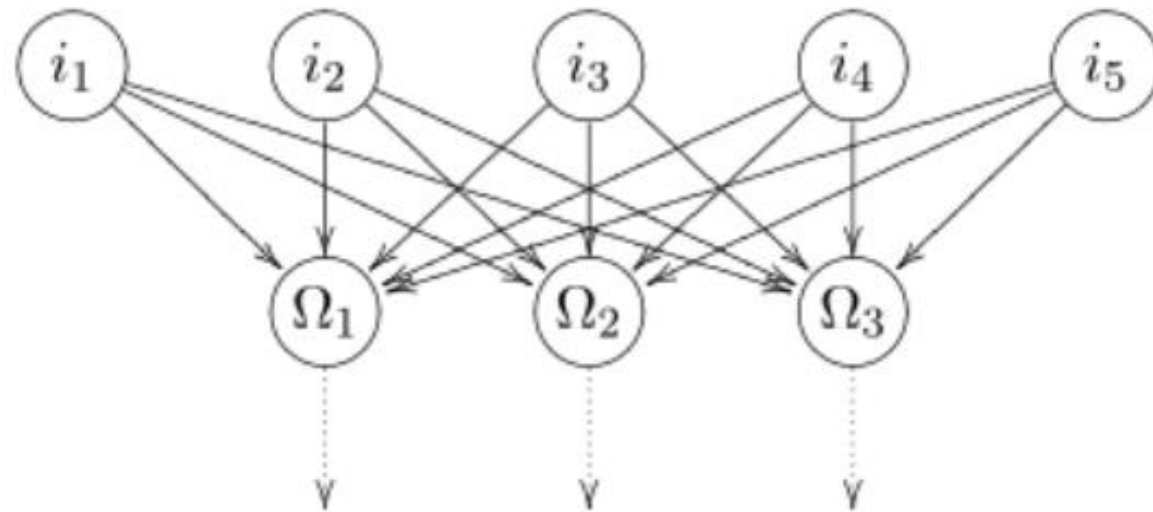


Neural network problems

Outline

1. Single-layer perceptron
 - a) Using perceptron learning algorithm
 - b) Using delta rule
2. Single-layer perceptron with multiple outputs

1. Single-layer perceptron



1. Single-layer perceptron

- Before using SLP, make sure the data is linearly separable
 - Visualize the data (not possible for more than 2 features)

1. Single-layer perceptron

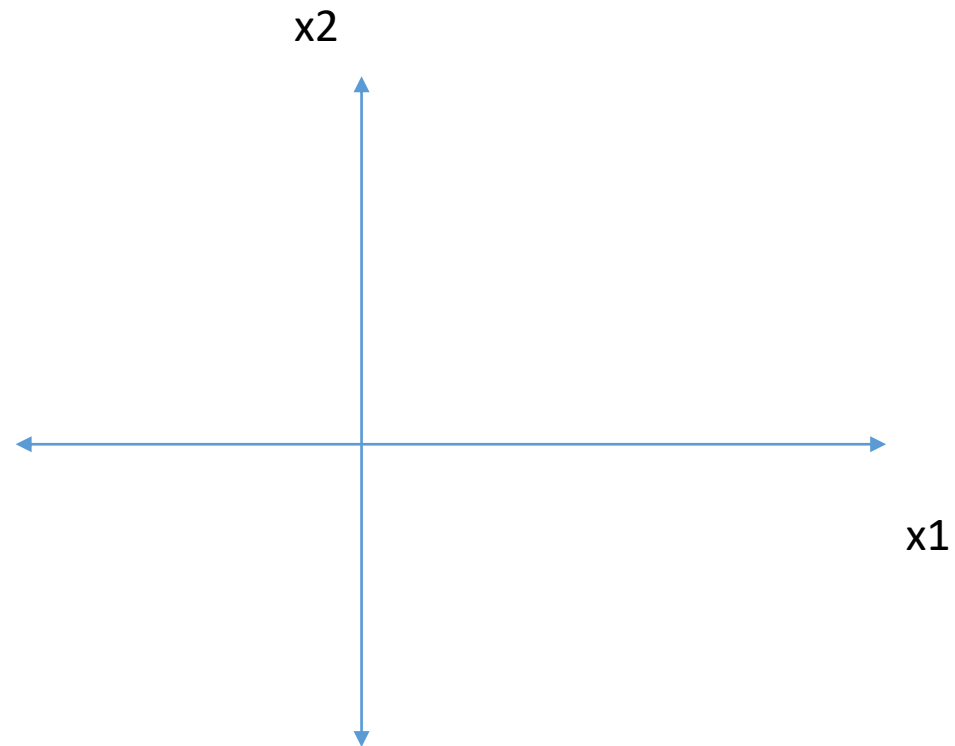
- Visualization example (2 features)

x1	x2	t
2	3	0
-3	3	1
3	4	0
-1	2	1
7	2	0
0	1	1
0	-2	1

1. Single-layer perceptron

- Visualization example (2 features)

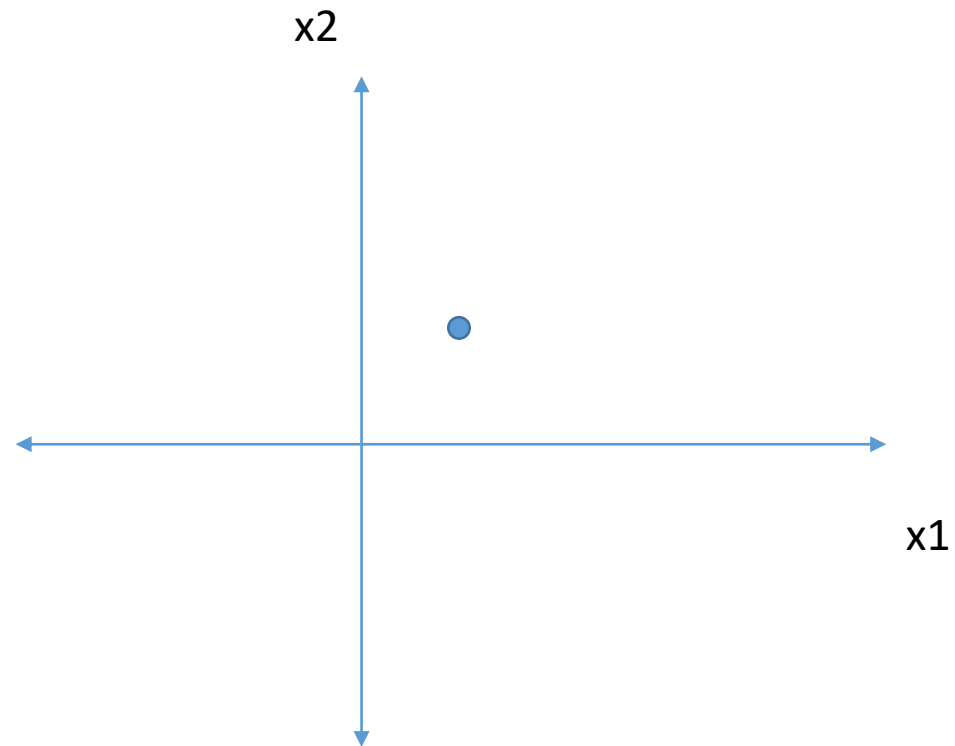
x1	x2	t
2	3	0
-3	3	1
3	4	0
-1	2	1
7	2	0
0	1	1
0	-2	1



1. Single-layer perceptron

- Visualization example (2 features)

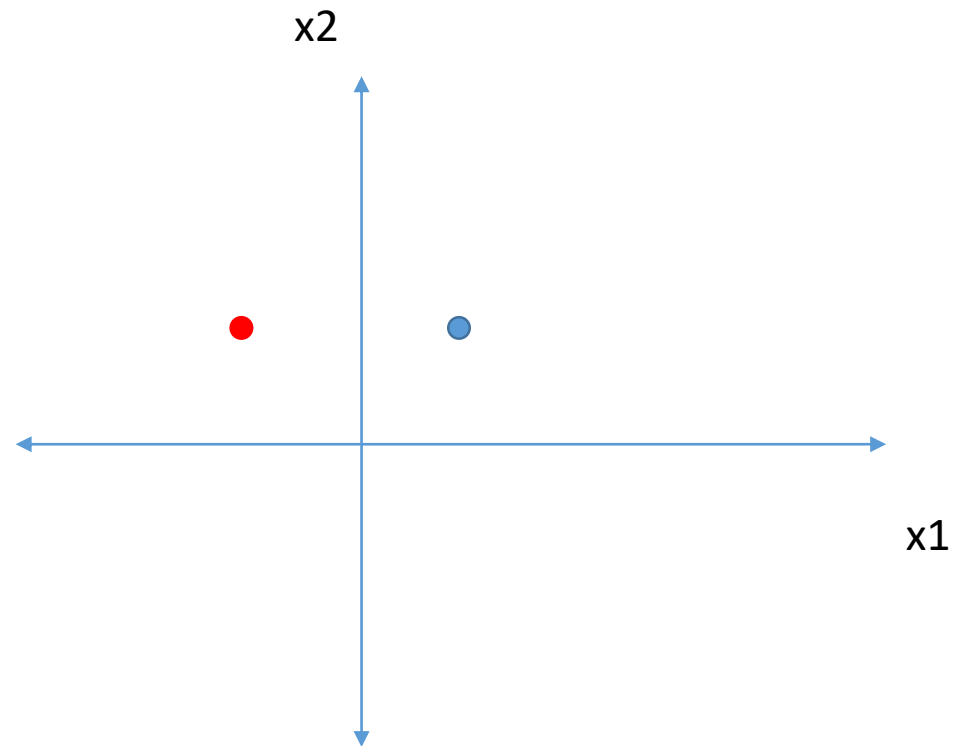
x1	x2	t
2	3	0
-3	3	1
3	4	0
-1	2	1
7	2	0
0	1	1
0	-2	1



1. Single-layer perceptron

- Visualization example (2 features)

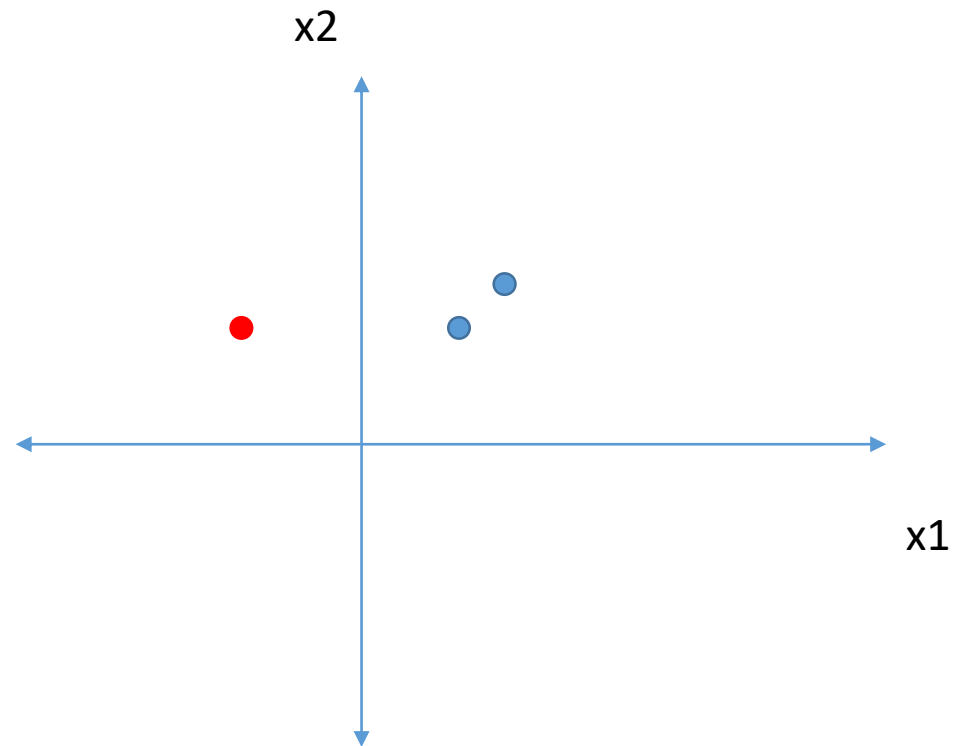
x1	x2	t
2	3	0
-3	3	1
3	4	0
-1	2	1
7	2	0
0	1	1
0	-2	1



1. Single-layer perceptron

- Visualization example (2 features)

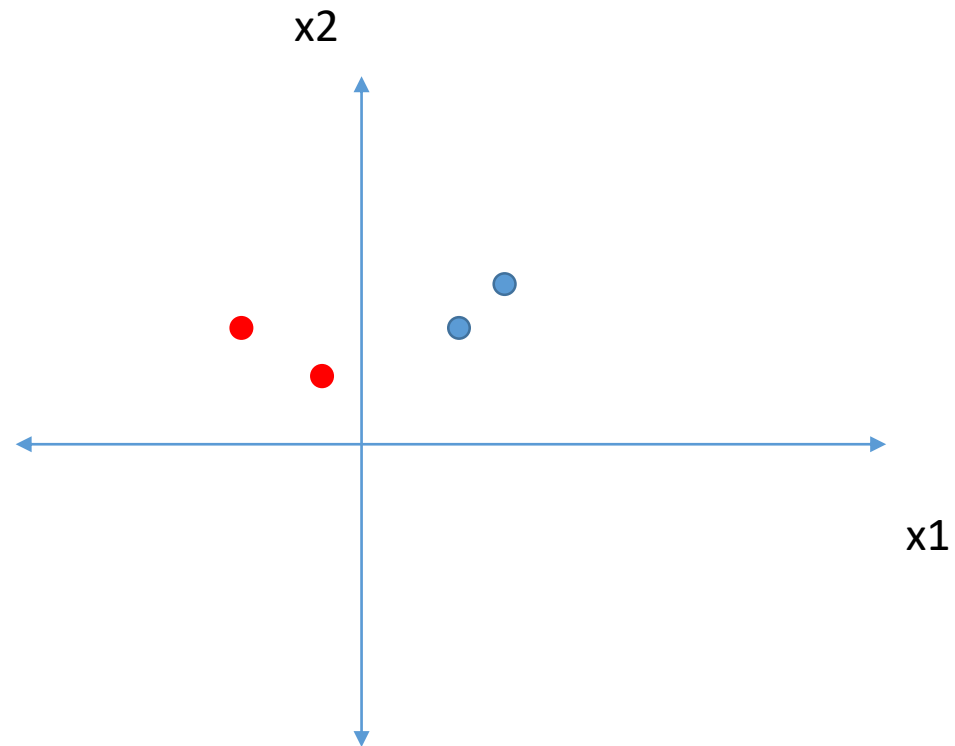
x1	x2	t
2	3	0
-3	3	1
3	4	0
-1	2	1
7	2	0
0	1	1
0	-2	1



1. Single-layer perceptron

- Visualization example (2 features)

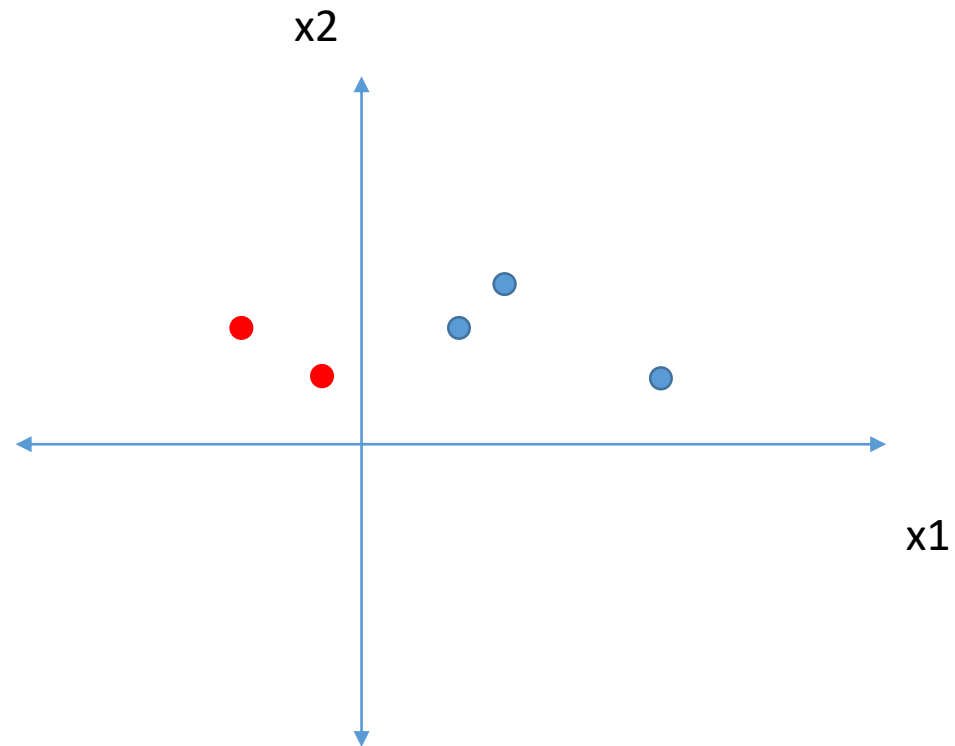
x1	x2	t
2	3	0
-3	3	1
3	4	0
-1	2	1
7	2	0
0	1	1
0	-2	1



1. Single-layer perceptron

- Visualization example (2 features)

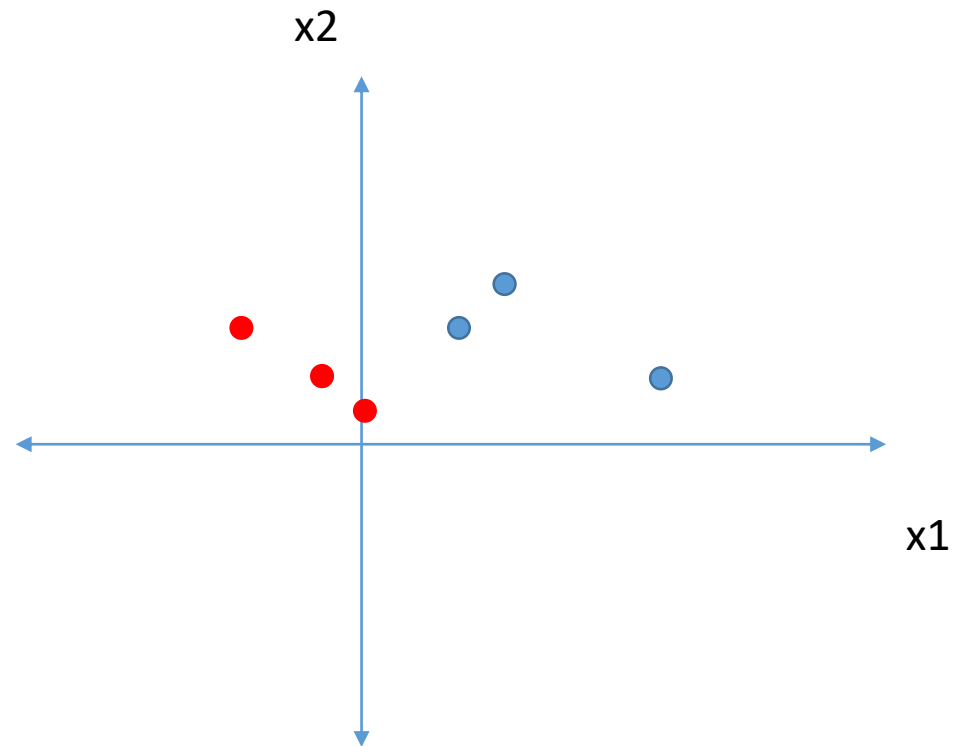
x1	x2	t
2	3	0
-3	3	1
3	4	0
-1	2	1
7	2	0
0	1	1
0	-2	1



1. Single-layer perceptron

- Visualization example (2 features)

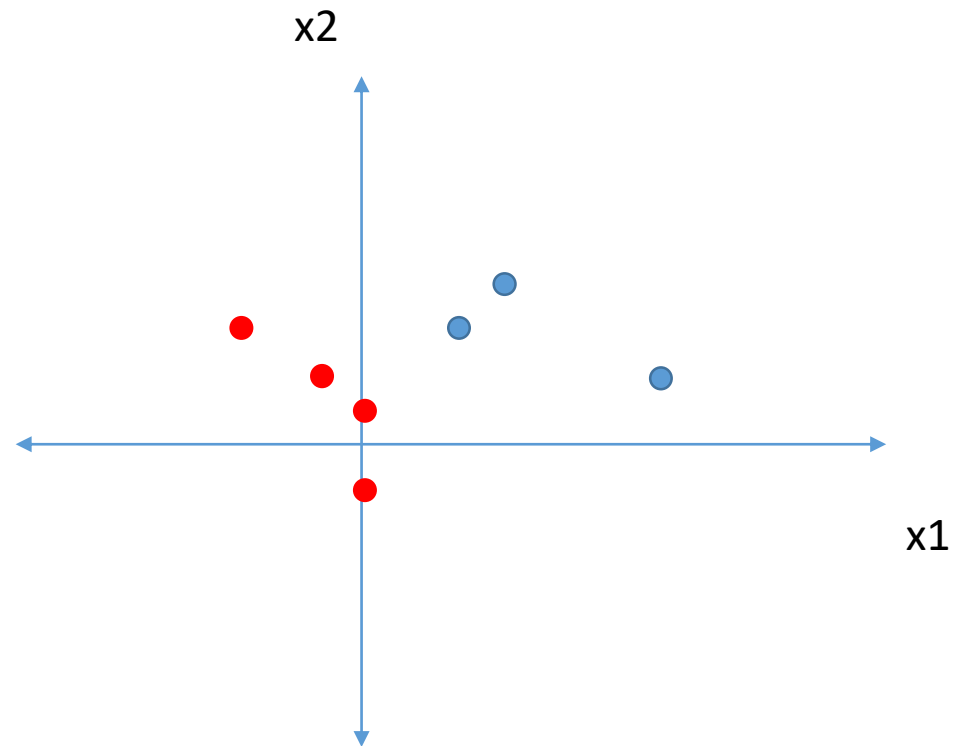
x1	x2	t
2	3	0
-3	3	1
3	4	0
-1	2	1
7	2	0
0	1	1
0	-2	1



1. Single-layer perceptron

- Visualization example (2 features)

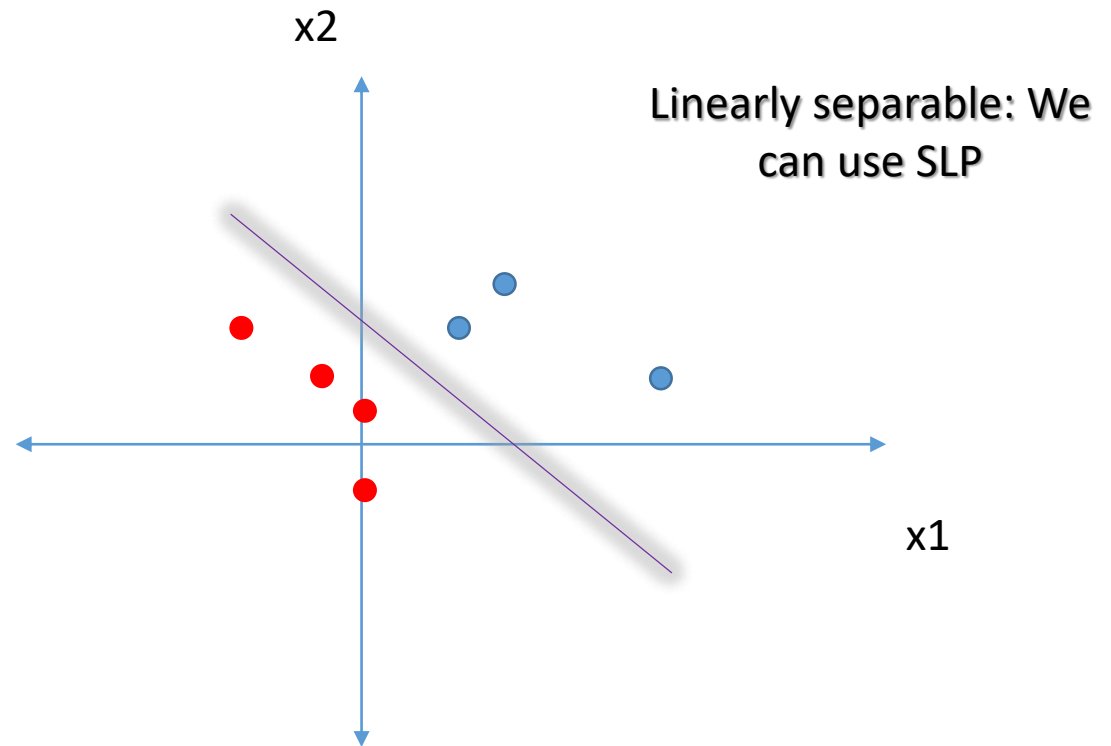
x1	x2	t
2	3	0
-3	3	1
3	4	0
-1	2	1
7	2	0
0	1	1
0	-2	1



1. Single-layer perceptron

- Visualization example (2 features)

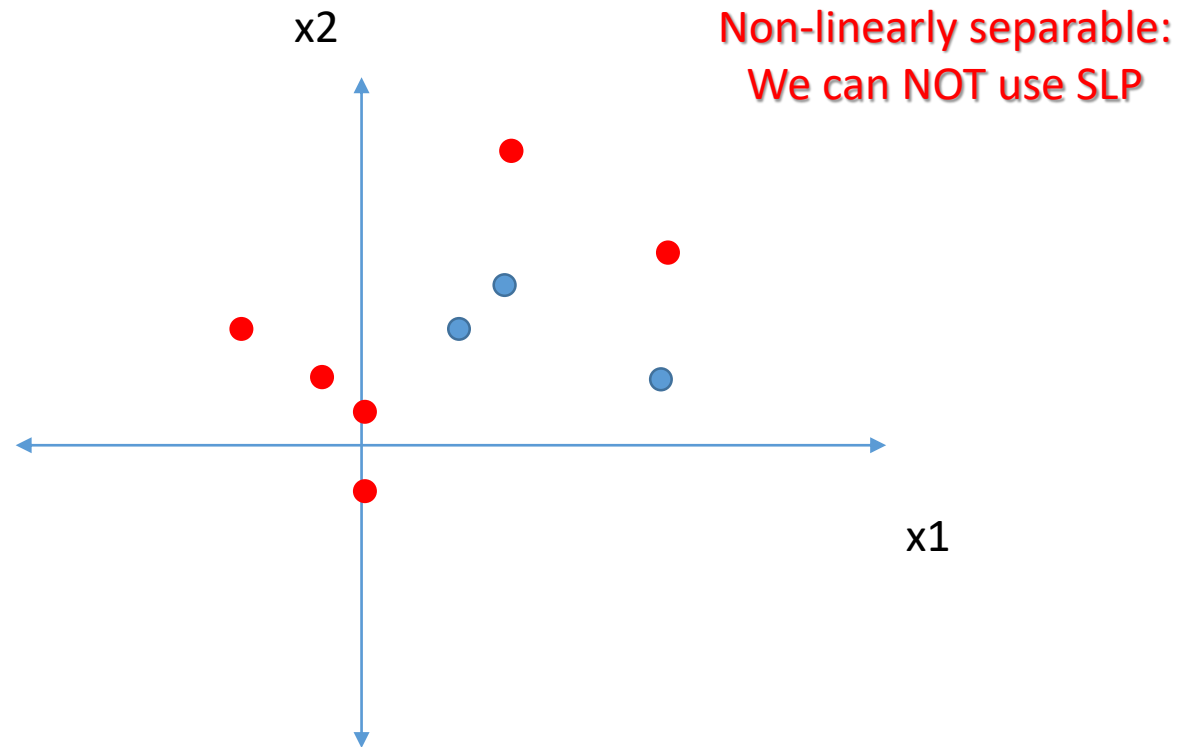
x1	x2	t
2	3	0
-3	3	1
3	4	0
-1	2	1
7	2	0
0	1	1
0	-2	1



1. Single-layer perceptron

- Visualization example (2 features)

x1	x2	t
2	3	0
-3	3	1
3	4	0
-1	2	1
7	2	0
0	1	1
0	-2	1
3	8	1
7	5	1



1. Single-layer perceptron

- Visualization example (1 feature)

x	t
1.1	0
2.8	1
-0.5	0
1.2	0
4	1
2.6	0
3.4	1

1. Single-layer perceptron

- Visualization example (1 feature)

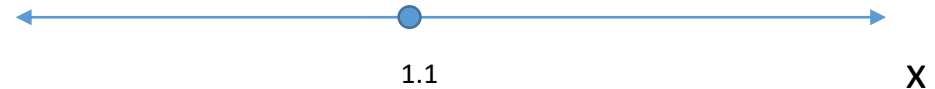
x	t
1.1	0
2.8	1
-0.5	0
1.2	0
4	1
2.6	0
3.4	1



1. Single-layer perceptron

- Visualization example (1 feature)

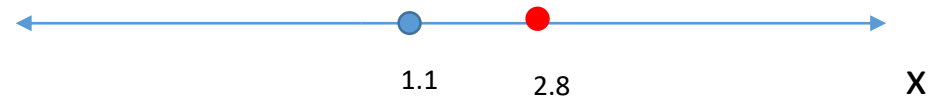
x	t
1.1	0
2.8	1
-0.5	0
1.2	0
4	1
2.6	0
3.4	1



1. Single-layer perceptron

- Visualization example (1 feature)

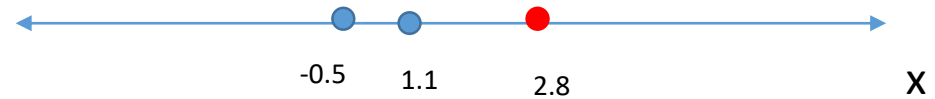
x	t
1.1	0
2.8	1
-0.5	0
1.2	0
4	1
2.6	0
3.4	1



1. Single-layer perceptron

- Visualization example (1 feature)

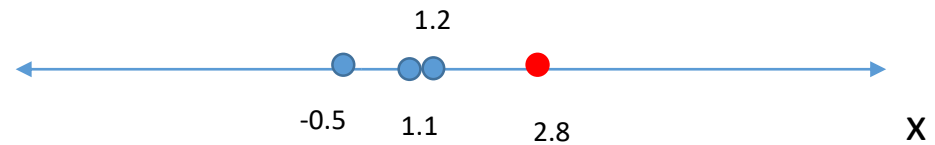
x	t
1.1	0
2.8	1
-0.5	0
1.2	0
4	1
2.6	0
3.4	1



1. Single-layer perceptron

- Visualization example (1 feature)

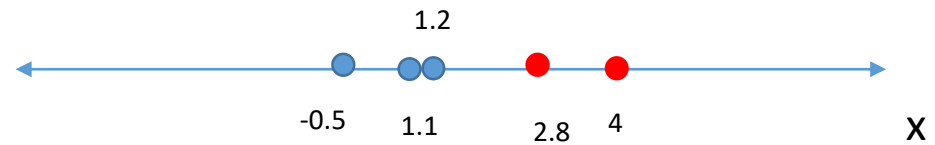
x	t
1.1	0
2.8	1
-0.5	0
1.2	0
4	1
2.6	0
3.4	1



1. Single-layer perceptron

- Visualization example (1 feature)

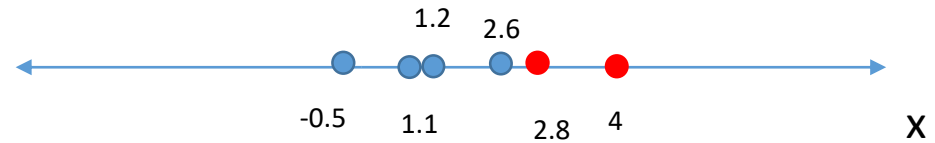
x	t
1.1	0
2.8	1
-0.5	0
1.2	0
4	1
2.6	0
3.4	1



1. Single-layer perceptron

- Visualization example (1 feature)

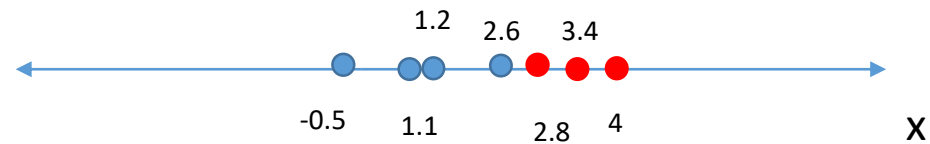
x	t
1.1	0
2.8	1
-0.5	0
1.2	0
4	1
2.6	0
3.4	1



1. Single-layer perceptron

- Visualization example (1 feature)

x	t
1.1	0
2.8	1
-0.5	0
1.2	0
4	1
2.6	0
3.4	1

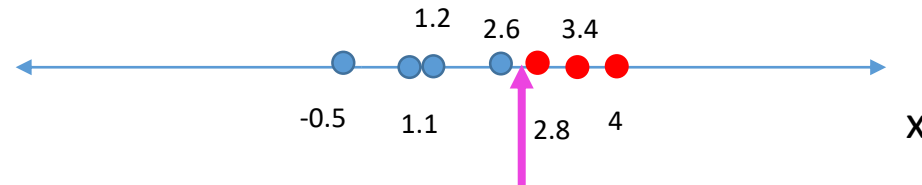


1. Single-layer perceptron

- Visualization example (1 feature)

x	t
1.1	0
2.8	1
-0.5	0
1.2	0
4	1
2.6	0
3.4	1

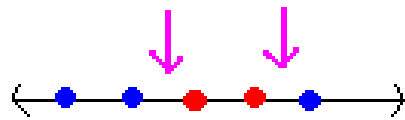
Note: in 1d, SLP is a point separator



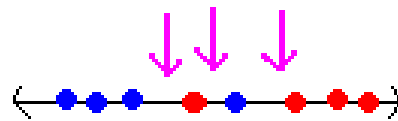
We can separate
the two classes
with one point:
We can use SLP

1. Single-layer perceptron

- Visualization example (1 feature)



Need at least 2 points:
Can't use SLP



Need at least 3 points:
Can't use SLP

1. a) Perceptron learning algorithm

```
1: while  $\exists p \in P$  and error too large do
2:   Input  $p$  into the network, calculate output  $y$   $\{P$  set of training patterns $\}$ 
3:   for all output neurons  $\Omega$  do
4:     if  $y_{\Omega} = t_{\Omega}$  then
5:       Output is okay, no correction of weights
6:     else
7:       if  $y_{\Omega} = 0$  then  $E = T - y = -1, +1$ 
8:         for all input neurons  $i$  do
9:            $w_{i,\Omega} := w_{i,\Omega} + o_i$   $\{\dots$ increase weight towards  $\Omega$  by  $o_i$  $\}$ 
10:        end for
11:      end if
12:      if  $y_{\Omega} = 1$  then
13:        for all input neurons  $i$  do
14:           $w_{i,\Omega} := w_{i,\Omega} - o_i$   $\{\dots$ decrease weight towards  $\Omega$  by  $o_i$  $\}$ 
15:        end for
16:      end if
17:    end if
18:  end for
19: end while
```

1. a) Perceptron learning algorithm

```
1: while  $\exists p \in P$  and error too large do
2:   Input  $p$  into the network, calculate output  $y$   $\{P$  set of training patterns $\}$ 
3:   for all output neurons  $\Omega$  do
4:     if  $y_{\Omega} = t_{\Omega}$  then
5:       Output is okay, no correction of weights
6:     else
7:       if  $y_{\Omega} = 0$  then
8:         for all input neurons  $i$  do
9:            $w_{i,\Omega} := w_{i,\Omega} + o_i$  {...increase weight towards  $\Omega$  by  $o_i$ }
10:        end for
11:      end if
12:      if  $y_{\Omega} = 1$  then
13:        for all input neurons  $i$  do
14:           $w_{i,\Omega} := w_{i,\Omega} - o_i$  {...decrease weight towards  $\Omega$  by  $o_i$ }
15:        end for
16:      end if
17:    end if
18:  end for
19: end while
```

1. a) Perceptron learning algorithm

```
1: while  $\exists p \in P$  and error too large do
2:   Input  $p$  into the network, calculate output  $y$   $\{P$  set of training patterns $\}$ 
3:   for all output neurons  $\Omega$  do
4:     if  $y_\Omega = t_\Omega$  then
5:       Output is okay, no correction of weights
6:     else
7:       if  $y_\Omega = 0$  then
8:         for all input neurons  $i$  do
9:            $w_{i,\Omega} := w_{i,\Omega} + o_i$  {...increase weight towards  $\Omega$  by  $o_i$ }
10:        end for
11:      end if
12:      if  $y_\Omega = 1$  then
13:        for all input neurons  $i$  do
14:            $w_{i,\Omega} := w_{i,\Omega} - o_i$  {...decrease weight towards  $\Omega$  by  $o_i$ }
15:        end for
16:      end if
17:    end if
18:  end for
19: end while
```

1. a) Perceptron learning algorithm

$$w_{i,\Omega} := w_{i,\Omega} + o_i$$

$$w_{i,\Omega} := w_{i,\Omega} - o_i$$

1. a) Perceptron learning algorithm

$$w_{i,\Omega} := w_{i,\Omega} + o_i$$

Step size depends on O_i .

Not controlled because O_i
can be large.

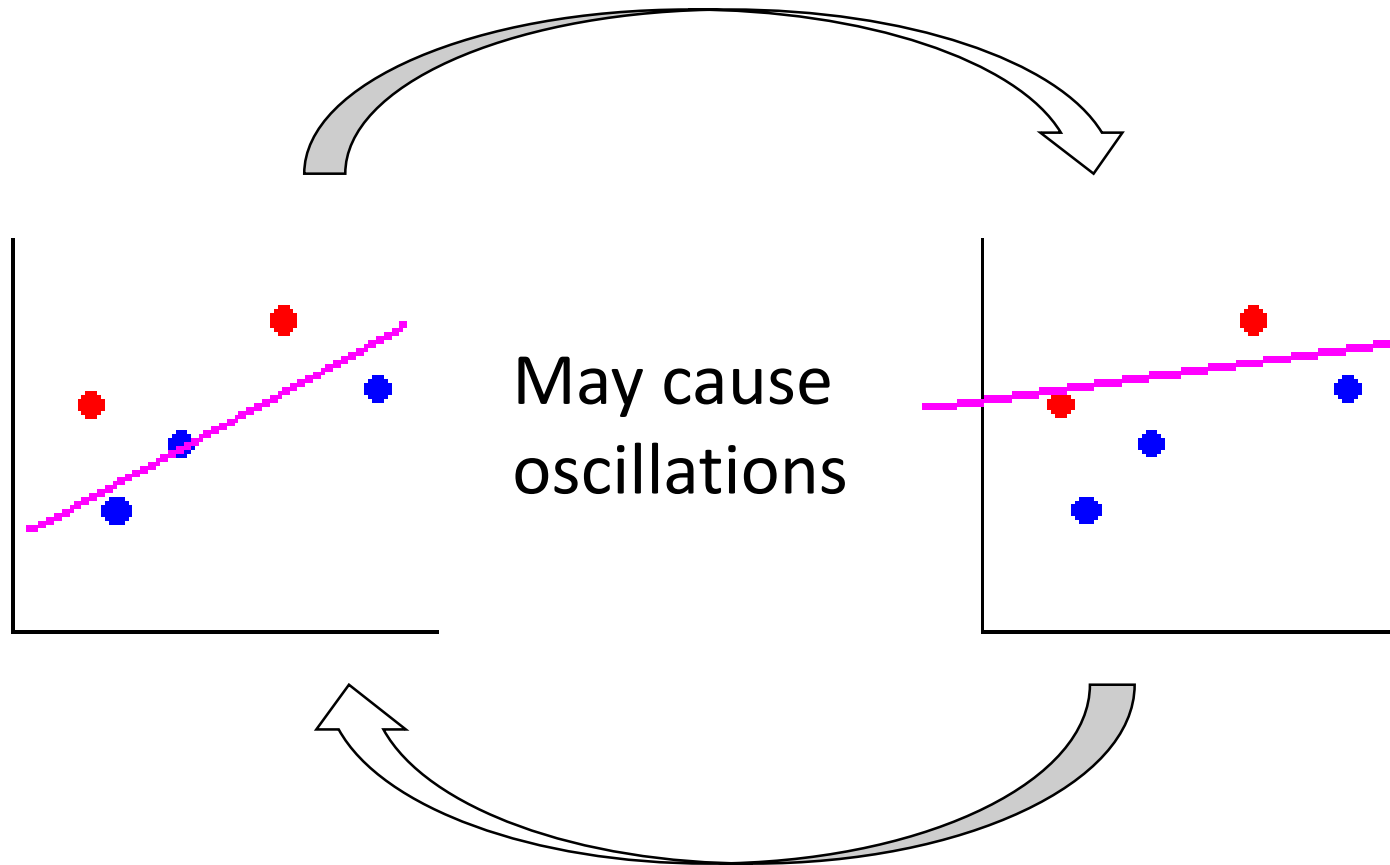
$$w_{i,\Omega} := w_{i,\Omega} - o_i$$

1. a) Perceptron learning algorithm

It assumes
the output
is either 0
or 1

```
1: while  $\exists p \in P$  and error too large do
2:   Input  $p$  into the network, calculate output  $y$   $\{P$  set of training patterns $\}$ 
3:   for all output neurons  $\Omega$  do
4:     if  $y_{\Omega} = t_{\Omega}$  then
5:       Output is okay, no correction of weights
6:     else
7:       if  $y_{\Omega} = 0$  then
8:         for all input neurons  $i$  do
9:            $w_{i,\Omega} := w_{i,\Omega} + o_i$   $\{\dots$ increase weight towards  $\Omega$  by  $o_i$  $\}$ 
10:        end for
11:      end if
12:      if  $y_{\Omega} = 1$  then
13:        for all input neurons  $i$  do
14:           $w_{i,\Omega} := w_{i,\Omega} - o_i$   $\{\dots$ decrease weight towards  $\Omega$  by  $o_i$  $\}$ 
15:        end for
16:      end if
17:    end if
18:  end for
19: end while
```


1. a) Perceptron learning algorithm



1. a) Perceptron learning algorithm example

x1	x2	t
0	0	0
0	1	1
1	0	1
1	1	1

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0							0
0	1							1
1	0							1
1	1							1



Add new columns

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1						0
0	1	1						1
1	0	1						1
1	1	1						1



Bias node always
produces 1

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2			0
0	1	1						1
1	0	1						1
1	1	1						1

Put initial weights (given)

If not given: assume random weights
(but not 0)

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2		0
0	1	1						1
1	0	1						1
1	1	1						1

Calculate net = $x1*w1 + x2*w2 + bias*w_bias$

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1						1
1	0	1						1
1	1	1						1

$$E = T - y = 0 - 0 = 0$$

Calculate $y =$

1 if $\text{net} \geq \text{threshold}$,
0 if $\text{net} < \text{threshold}$

Threshold should be given. If not, assume random threshold

Here we assume $\text{threshold} = 0.1 \rightarrow \text{net} < \text{threshold}$

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1						1
1	0	1						1
1	1	1						1

$y = t$? yes

Weights will not be changed

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2			1
1	0	1						1
1	1	1						1

Use the same weights for next pattern

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0		1
1	0	1						1
1	1	1						1

Calculate net = $x1*w1 + x2*w2 + bias*w_bias$

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1						1
1	1	1						1

$\text{net} < 0.1 \rightarrow y = 0$

$E = T - y = 1 - 0 = 1$
 $w_{\text{new}} = w_{\text{old}} + X$

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1						1
1	1	1						1

$y \neq t$

We need to change weights

$y = 0$ we want $y = 1$ increase weights

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1	0.1	1.2	0.8			1
1	1	1						1

$w1 := w1 + x1$
 $w2 := w1 + x1$
 $w_bias := w_bias + bias$

$E = T - y = 1 - 0 = 1$
 $W_{new} = w_{old} + x * E$

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1	0.1	1.2	0.8	0.9		1
1	1	1						1

Calculate net = $x1*w1 + x2*w2 + bias*w_bias$

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1	0.1	1.2	0.8	0.9	1	1
1	1	1						1

$\text{net} \geq 0.1 \quad \rightarrow \quad y = 1$

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1	0.1	1.2	0.8	0.9	1	1
1	1	1	0.1	1.2	0.8			1

$$y = t$$

Don't change weights

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1	0.1	1.2	0.8	0.9	1	1
1	1	1	0.1	1.2	0.8	2.1		1

Calculate net = $x1*w1 + x2*w2 + bias*w_bias$

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1	0.1	1.2	0.8	0.9	1	1
1	1	1	0.1	1.2	0.8	2.1	1	1

$\text{net} \geq 0.1 \quad \rightarrow \quad y = 1$

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1	0.1	1.2	0.8	0.9	1	1
1	1	1	0.1	1.2	0.8	2.1	1	1
Weights for next epoch			0.1	1.2	0.8			

$$y = t$$

Don't change weights

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1	0.1	1.2	0.8	0.9	1	1
1	1	1	0.1	1.2	0.8	2.1	1	1
Weights for next epoch			0.1	1.2	0.8			

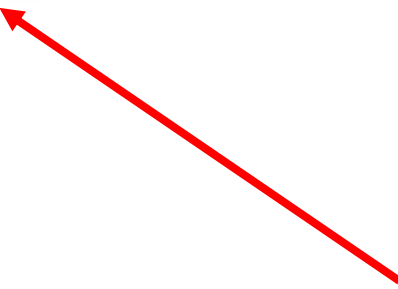
1 Epoch complete:

But we still have 1 error

We need to run another epoch

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1	0.1	1.2	0.8	0.9	1	1
1	1	1	0.1	1.2	0.8	2.1	1	1
Weights for next epoch			0.1	1.2	0.8			



Use these as
initial weights
for next epoch

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	1.2	0.8			0
0	1	1						1
1	0	1						1
1	1	1						1

New epoch with initial weights from previous slide

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	1.2	0.8	0.8	1	0
0	1	1						1
1	0	1						1
1	1	1						1

Calculate net and y

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	1.2	0.8	0.8	1	0
0	1	1	0.1	1.2	-0.2			1
1	0	1						1
1	1	1						1

$y \neq t$ $y = 1$ and we want $y = 0$

Decrease weights:

$w1 \quad := w1 - x1$

$w2 \quad := w1 - x1$

$w_bias := w_bias - bias$

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	1.2	0.8	0.8	1	0
0	1	1	0.1	1.2	-0.2	1	1	1
1	0	1						1
1	1	1						1

Calculate net and y

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	1.2	0.8	0.8	1	0
0	1	1	0.1	1.2	-0.2	1	1	1
1	0	1	0.1	1.2	-0.2			1
1	1	1						1

$$y = t$$

Don't change weights

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	1.2	0.8	0.8	1	0
0	1	1	0.1	1.2	-0.2	1	1	1
1	0	1	0.1	1.2	-0.2	-0.1	0	1
1	1	1						1

Calculate net and y

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	1.2	0.8	0.8	1	0
0	1	1	0.1	1.2	-0.2	1	1	1
1	0	1	0.1	1.2	-0.2	-0.1	0	1
1	1	1	1.1	1.2	0.8			1

$y = 0$ and we want $y = 1$

Increase weights:

$w1 := w1 + x1$

$w2 := w1 + x1$

$w_bias := w_bias + bias$

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	1.2	0.8	0.8	1	0
0	1	1	0.1	1.2	-0.2	1	1	1
1	0	1	0.1	1.2	-0.2	-0.1	0	1
1	1	1	1.1	1.2	0.8	3.1	1	1

Calculate net and y

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	1.2	0.8	0.8	1	0
0	1	1	0.1	1.2	-0.2	1	1	1
1	0	1	0.1	1.2	-0.2	-0.1	0	1
1	1	1	1.1	1.2	0.8	3.1	1	1
Weights for next epoch			1.1	1.2	0.8			

$$y = t$$

Don't change weights

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	0.1	1.2	0.8	0.8	1	0
0	1	1	0.1	1.2	-0.2	1	1	1
1	0	1	0.1	1.2	-0.2	-0.1	0	1
1	1	1	1.1	1.2	0.8	3.1	1	1
Weights for next epoch			1.1	1.2	0.8			

Second epoch done

We still have 2 errors

We need to run another epoch

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	0.8			0
0	1	1						1
1	0	1						1
1	1	1						1

Starting third epoch

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	0.8	0.8	1	0
0	1	1						1
1	0	1						1
1	1	1						1

Calculate net and y

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	0.8	0.8	1	0
0	1	1	1.1	1.2	-0.2			1
1	0	1						1
1	1	1						1

Decrease weights

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	0.8	0.8	1	0
0	1	1	1.1	1.2	-0.2	1	1	1
1	0	1						1
1	1	1						1

Calculate net and y

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	0.8	0.8	1	0
0	1	1	1.1	1.2	-0.2	1	1	1
1	0	1	1.1	1.2	-0.2			1
1	1	1						1

Don't change weights

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	0.8	0.8	1	0
0	1	1	1.1	1.2	-0.2	1	1	1
1	0	1	1.1	1.2	-0.2	0.9	1	1
1	1	1						1

Calculate net and y

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	0.8	0.8	1	0
0	1	1	1.1	1.2	-0.2	1	1	1
1	0	1	1.1	1.2	-0.2	0.9	1	1
1	1	1	1.1	1.2	-0.2			1

Don't change weights

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	0.8	0.8	1	0
0	1	1	1.1	1.2	-0.2	1	1	1
1	0	1	1.1	1.2	-0.2	0.9	1	1
1	1	1	1.1	1.2	-0.2	2.1	1	1

Calculate net and y

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	0.8	0.8	1	0
0	1	1	1.1	1.2	-0.2	1	1	1
1	0	1	1.1	1.2	-0.2	0.9	1	1
1	1	1	1.1	1.2	-0.2	2.1	1	1
Weights for next epoch			1.1	1.2	-0.2			

Don't change weights

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	0.8	0.8	1	0
0	1	1	1.1	1.2	-0.2	1	1	1
1	0	1	1.1	1.2	-0.2	0.9	1	1
1	1	1	1.1	1.2	-0.2	2.1	1	1
Weights for next epoch			1.1	1.2	-0.2			

We still have one error

We need to run another epoch

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	-0.2			0
0	1	1						1
1	0	1						1
1	1	1						1

Fourth epoch

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	-0.2	-0.2	0	0
0	1	1						1
1	0	1						1
1	1	1						1

Calculate net and y

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	-0.2	-0.2	0	0
0	1	1	1.1	1.2	-0.2			1
1	0	1						1
1	1	1						1

Don't change weights

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	-0.2	-0.2	0	0
0	1	1	1.1	1.2	-0.2	1	1	1
1	0	1						1
1	1	1						1

Calculate net and y

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	-0.2	-0.2	0	0
0	1	1	1.1	1.2	-0.2	1	1	1
1	0	1	1.1	1.2	-0.2			1
1	1	1						1

Don't change weights

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	-0.2	-0.2	0	0
0	1	1	1.1	1.2	-0.2	1	1	1
1	0	1	1.1	1.2	-0.2	0.9	1	1
1	1	1						1

Calculate net and y

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	-0.2	-0.2	0	0
0	1	1	1.1	1.2	-0.2	1	1	1
1	0	1	1.1	1.2	-0.2	0.9	1	1
1	1	1	1.1	1.2	-0.2			1

Don't change weights

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	-0.2	-0.2	0	0
0	1	1	1.1	1.2	-0.2	1	1	1
1	0	1	1.1	1.2	-0.2	0.9	1	1
1	1	1	1.1	1.2	-0.2	2.1	1	1

Calculate net and y

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	-0.2	-0.2	0	0
0	1	1	1.1	1.2	-0.2	1	1	1
1	0	1	1.1	1.2	-0.2	0.9	1	1
1	1	1	1.1	1.2	-0.2	2.1	1	1
Weights for next epoch			1.1	1.2	-0.2			

Don't change weights

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	-0.2	-0.2	0	0
0	1	1	1.1	1.2	-0.2	1	1	1
1	0	1	1.1	1.2	-0.2	0.9	1	1
1	1	1	1.1	1.2	-0.2	2.1	1	1
Weights for next epoch			1.1	1.2	-0.2			

Fourth epoch done

No errors



Stop training

y =

1. a) Perceptron learning algorithm

x1	x2	bias	w1	w2	w_bias	net	y	t
0	0	1	1.1	1.2	-0.2	-0.2	0	0
0	1	1	1.1	1.2	-0.2	1	1	1
1	0	1	1.1	1.2	-0.2	0.9	1	1
1	1	1	1.1	1.2	-0.2	2.1	1	1
Weights for next epoch			1.1	1.2	-0.2			

Final weights



1. b) SLP using delta rule

- Same as the previous example. Just updating weights is different

$$w_{i,\Omega} := w_{i,\Omega} + \eta o_i (t_\Omega - y_\Omega)$$

- For previous example:

- $w_1 := w_1 + \eta * x_1 * (t - y)$

- $w_2 := w_2 + \eta * x_2 * (t - y)$

- $w_bias := w_bias + \eta * bias * (t - y)$

1. b) SLP using delta rule

- Same as the previous example. Just updating weights is different

$$w_{i,\Omega} := w_{i,\Omega} + \eta o_i (t_\Omega - y_\Omega)$$

- For previous example:

- $w_1 := w_1 + \eta * x_1 * (t - y)$

- $w_2 := w_2 + \eta * x_2 * (t - y)$

- $w_bias := w_bias + \eta * \text{bias} * (t - y)$

The term “bias”
always equals 1
(can be omitted)

1. b) SLP using delta rule

- Same as the previous example. Just updating weights is different

$$w_{i,\Omega} := w_{i,\Omega} + \eta o_i (t_{\Omega} - y_{\Omega})$$

- For previous example:

- $w_1 := w_1 + \eta * x_1 * (t - y)$
- $w_2 := w_2 + \eta * x_2 * (t - y)$
- $w_bias := w_bias + \eta * bias * (t - y)$

This is the learning rate (a given constant). If not given, assume a value between 0.01 and 0.9

1. b) SLP using delta rule

- Same as the previous example. Just updating weights is different

$$w_{i,\Omega} := w_{i,\Omega} + \eta o_i (t_{\Omega} - y_{\Omega})$$

- For previous example:

- $w_1 := w_1 + \eta * x_1 * (t - y)$
- $w_2 := w_2 + \eta * x_2 * (t - y)$
- $w_bias := w_bias + \eta * bias * (t - y)$

We always add
(even if $y > t$)

But how do we
decrease weights?

1. b) SLP using delta rule

- Same as the previous example. Just updating weights is different

$$w_{i,\Omega} := w_{i,\Omega} + \eta o_i (t_{\Omega} - y_{\Omega})$$

- For previous example:

- $w_1 := w_1 + \eta * x_1 * (t - y)$
- $w_2 := w_2 + \eta * x_2 * (t - y)$
- $w_bias := w_bias + \eta * bias * (t - y)$

If $y > t$, this term
will be negative,
causing weights to
be decreased

1. b) SLP using delta rule

x1	x2	bias	w1	w2	wb	net	y	t
0	0	1	0.1	0.2	-0.2			0
0	1	1						1
1	0	1						1
1	1	1						1

Same example using delta rule

Assume learning rate = 0.1

1. b) SLP using delta rule

x1	x2	bias	w1	w2	wb	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1						1
1	0	1						1
1	1	1						1

Calculating net and y is not different

1. b) SLP using delta rule

x1	x2	bias	w1	w2	wb	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2			1
1	0	1						1
1	1	1						1

If we try to update weights: (even though $y = t$)

$$w1 := w1 + 0.1 * x1 * (t - y)$$

$$w2 := w2 + 0.1 * x2 * (t - y)$$

$$wb := wb + 0.1 * bias * (t - y)$$

$(t - y) = 0$ so the weights will not be changed

1. b) SLP using delta rule

x1	x2	bias	w1	w2	wb	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1						1
1	1	1						1

Calculate y and net

1. b) SLP using delta rule

x1	x2	bias	w1	w2	wb	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1	0.1	0.3	-0.1			1
1	1	1						1

learning rate = 0.1

update weights:

$$\begin{array}{llll} w1 := w1 + 0.1 * x1 * (t - y) & \rightarrow & w1 := 0.1 + 0.1 * 0 * 1 & \rightarrow 0.1 \\ w2 := w2 + 0.1 * x2 * (t - y) & \rightarrow & w2 := 0.2 + 0.1 * 1 * 1 & \rightarrow 0.3 \\ wb := wb + 0.1 * bias * (t - y) & \rightarrow & wb := -0.2 + 0.1 * 1 * 1 & \rightarrow -0.1 \end{array}$$

1. b) SLP using delta rule

x1	x2	bias	w1	w2	wb	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1	0.1	0.3	-0.1	0	0	1
1	1	1						1

Calculate net and y

1. b) SLP using delta rule

x1	x2	bias	w1	w2	wb	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1	0.1	0.3	-0.1	0	0	1
1	1	1	0.2	0.3	0			1

update weights:

$$\begin{aligned}w1 &:= w1 + 0.1 * x1 * (t - y) &\rightarrow & 0.1 + 0.1 * 1 * 1 &\rightarrow 0.2 \\w2 &:= w2 + 0.1 * x2 * (t - y) &\rightarrow & 0.3 + 0.1 * 0 * 1 &\rightarrow 0.3 \\wb &:= wb + 0.1 * bias * (t - y) &\rightarrow & -0.1 + 0.1 * 1 * 1 &\rightarrow 0\end{aligned}$$

1. b) SLP using delta rule

x1	x2	bias	w1	w2	wb	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1	0.1	0.3	-0.1	0	0	1
1	1	1	0.2	0.3	0	0.5	1	1

Calculate net and y

1. b) SLP using delta rule

x1	x2	bias	w1	w2	wb	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1	0.1	0.3	-0.1	0	0	1
1	1	1	0.2	0.3	0	0.5	1	1
Weights for next epoch:			0.2	0.3	0			

Weights will not be changed

1. b) SLP using delta rule

x1	x2	bias	w1	w2	wb	net	y	t
0	0	1	0.1	0.2	-0.2	-0.2	0	0
0	1	1	0.1	0.2	-0.2	0	0	1
1	0	1	0.1	0.3	-0.1	0	0	1
1	1	1	0.2	0.3	0	0.5	1	1
Weights for next epoch:			0.2	0.3	0			

First epoch done

We have 2 errors

We need to run another epoch

1. b) SLP using delta rule

x1	x2	bias	w1	w2	wb	net	y	t
0	0	1	0.2	0.3	0			0
0	1	1						1
1	0	1						1
1	1	1						1

Second epoch

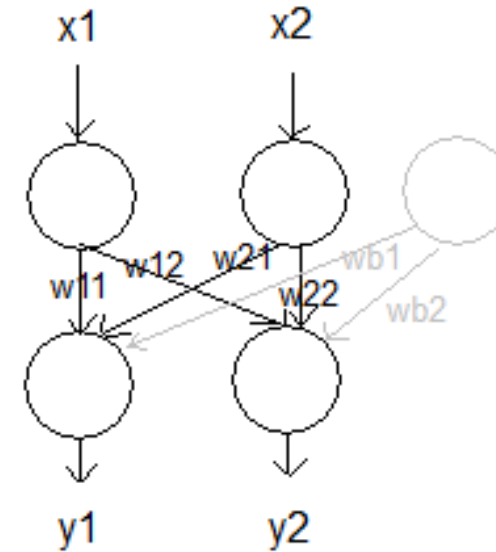
1. b) SLP using delta rule

x1	x2	bias	w1	w2	wb	net	y	t
0	0	1	0.2	0.3	0	0	0	0
0	1	1	0.2	0.3	0	0.3	1	1
1	0	1	0.2	0.3	0	0.2	1	1
1	1	1	0.2	0.3	0	0.5	1	1
Weights for next epoch:			0.2	0.3	0			

Second epoch has no errors → stop training

2. SLP with multiple outputs

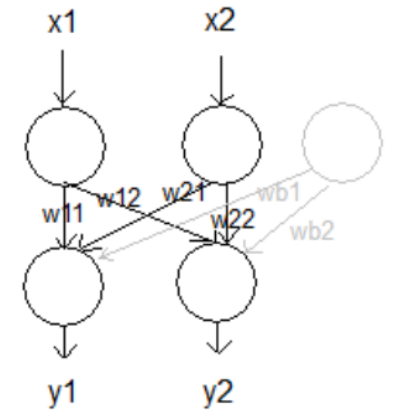
x1	x2	t1	t2
0	0	0	0
0	1	1	0
1	0	1	0
1	1	1	1



The two output neurons can have different threshold values

2. SLP with multiple outputs

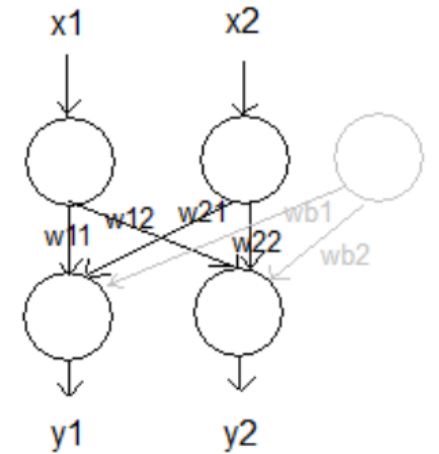
x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0									0	0
0	1									1	0
1	0									1	0
1	1									1	1



2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1			0	0
0	1									1	0
1	0									1	0
1	1									1	1

Initial weights



2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0		0	0
0	1									1	0
1	0									1	0
1	1									1	1

$$\begin{aligned}\text{net1} &= w_{11} * x_1 + w_{21} * x_2 + b_1 \\ \text{net1} &= 0.1 * 0 + 0.2 * 0 - 0.2 = -0.2\end{aligned}$$

Assume threshold1 = 0.1,
threshold2 = 1

net1 >= 0.1 ? → y = 1
else? → y = 0

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0	1	0	0
0	1									1	0
1	0									1	0
1	1									1	1

$$\text{net2} = w_{12} * x_1 + w_{22} * x_2 + b_2$$

$$\text{net2} = 0.2 * 0 + 0.3 * 0 + 1 = 1$$

net2 >= 1 ? $\rightarrow y = 1$

else? $\rightarrow y = 0$

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0	1	0	0
0	1	0.1	0.2	-0.2						1	0
1	0									1	0
1	1									1	1

y1 is OK

don't change its weights

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0	1	0	0
0	1	0.1	0.2	-0.2						1	0
1	0									1	0
1	1									1	1

y2 is wrong

update its weights:

(We can either use perceptron learning algorithm or delta rule)

Assume we are using delta rule, $\eta = 0.1$

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0	1	0	0
0	1	0.1	0.2	-0.2	0.2	0.3	0.9			1	0
1	0									1	0
1	1									1	1

$$\begin{array}{llll} w12 := w12 + 0.1 * x1 * (t2 - y2) & \rightarrow & 0.2 + 0.1 * 0 * -1 & \rightarrow 0.2 \\ w22 := w22 + 0.1 * x2 * (t2 - y2) & \rightarrow & 0.3 + 0.1 * 0 * -1 & \rightarrow 0.3 \\ wb2 := wb2 + 0.1 * (t2 - y2) & \rightarrow & 1 + 0.1 * -1 & \rightarrow 0.9 \end{array}$$

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0	1	0	0
0	1	0.1	0.2	-0.2	0.2	0.3	0.9	0		1	0
1	0									1	0
1	1									1	1

calculate net1, y1

$$\text{net1} = 0.1 * 0 + 0.2 * 1 - 0.2 = 0$$

$$\text{net1} < 0.1 \rightarrow y1 = 0$$

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0	1	0	0
0	1	0.1	0.2	-0.2	0.2	0.3	0.9	0	1	1	0
1	0									1	0
1	1									1	1

calculate net2, y2

$$\text{net2} = 0.2 * 0 + 0.3 * 1 + 0.9 = 1.2 \quad \text{net2} \geq 1 \rightarrow y2 = 1$$

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0	1	0	0
0	1	0.1	0.2	-0.2	0.2	0.3	0.9	0	1	1	0
1	0	0.1	0.3	-0.1						1	0
1	1									1	1

Update weights of y1

$$w11 = 0.1 + 0.1 * 0 * (1 - 0) = 0.1$$

$$w21 = 0.2 + 0.1 * 1 * (1 - 0) = 0.3$$

$$wb1 = -0.2 + 0.1 * (1 - 0) = -0.1$$

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0	1	0	0
0	1	0.1	0.2	-0.2	0.2	0.3	0.9	0	1	1	0
1	0	0.1	0.3	-0.1	0.2	0.2	0.8			1	0
1	1									1	1

Update weights of y2

$$w12 = 0.2 + 0.1 * 0 * (0 - 1) = 0.2$$

$$w22 = 0.3 + 0.1 * 1 * (0 - 1) = 0.2$$

$$wb2 = 0.9 + 0.1 * (0 - 1) = 0.8$$

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0	1	0	0
0	1	0.1	0.2	-0.2	0.2	0.3	0.9	0	1	1	0
1	0	0.1	0.3	-0.1	0.2	0.2	0.8	0		1	0
1	1									1	1

Calculate net1 and y1

$$\text{net1} = 0.1 * 1 + 0.3 * 0 - 0.1 = 0 \quad \rightarrow \quad y1 = 0$$

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0	1	0	0
0	1	0.1	0.2	-0.2	0.2	0.3	0.9	0	1	1	0
1	0	0.1	0.3	-0.1	0.2	0.2	0.8	0	1	1	0
1	1									1	1

Calculate net2 and y2

$$\text{net2} = 0.2 * 1 + 0.2 * 0 + 0.8 = 1 \quad \rightarrow \quad y2 = 1$$

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0	1	0	0
0	1	0.1	0.2	-0.2	0.2	0.3	0.9	0	1	1	0
1	0	0.1	0.3	-0.1	0.2	0.2	0.8	0	1	1	0
1	1	0.2	0.3	0						1	1

Update weights of y1

$$w11 = 0.1 + 0.1 * 1 * (1 - 0) = 0.2$$

$$w21 = 0.3 + 0.1 * 0 * (1 - 0) = 0.3$$

$$wb1 = -0.1 + 0.1 * (1 - 0) = 0$$

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0	1	0	0
0	1	0.1	0.2	-0.2	0.2	0.3	0.9	0	1	1	0
1	0	0.1	0.3	-0.1	0.2	0.2	0.8	0	1	1	0
1	1	0.2	0.3	0	0.1	0.2	0.7			1	1

Update weights of y2

$$w12 = 0.2 + 0.1 * 1 * (0 - 1) = 0.1$$

$$w22 = 0.2 + 0.1 * 0 * (0 - 1) = 0.2$$

$$wb2 = 0.8 + 0.1 * (0 - 1) = 0.7$$

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0	1	0	0
0	1	0.1	0.2	-0.2	0.2	0.3	0.9	0	1	1	0
1	0	0.1	0.3	-0.1	0.2	0.2	0.8	0	1	1	0
1	1	0.2	0.3	0	0.1	0.2	0.7	1		1	1

Calculate net1 and y1

$$\text{net1} = 0.2 * 1 + 0.3 * 1 + 0 = 0.5 \quad \rightarrow y1 = 1$$

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0	1	0	0
0	1	0.1	0.2	-0.2	0.2	0.3	0.9	0	1	1	0
1	0	0.1	0.3	-0.1	0.2	0.2	0.8	0	1	1	0
1	1	0.2	0.3	0	0.1	0.2	0.7	1	1	1	1

Calculate net2 and y2

$$\text{net2} = 0.1 * 1 + 0.2 * 0.7 + 0 = 1 \quad \rightarrow y2 = 1$$

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0	1	0	0
0	1	0.1	0.2	-0.2	0.2	0.3	0.9	0	1	1	0
1	0	0.1	0.3	-0.1	0.2	0.2	0.8	0	1	1	0
1	1	0.2	0.3	0	0.1	0.2	0.7	1	1	1	1
Next weights		0.2	0.3	0	0.1	0.2	0.7				

both y1 and y2 are OK

Don't update weights

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.1	0.2	-0.2	0.2	0.3	1	0	1	0	0
0	1	0.1	0.2	-0.2	0.2	0.3	0.9	0	1	1	0
1	0	0.1	0.3	-0.1	0.2	0.2	0.8	0	1	1	0
1	1	0.2	0.3	0	0.1	0.2	0.7	1	1	1	1
Next weights		0.2	0.3	0	0.1	0.2	0.7				

We need to run another epoch

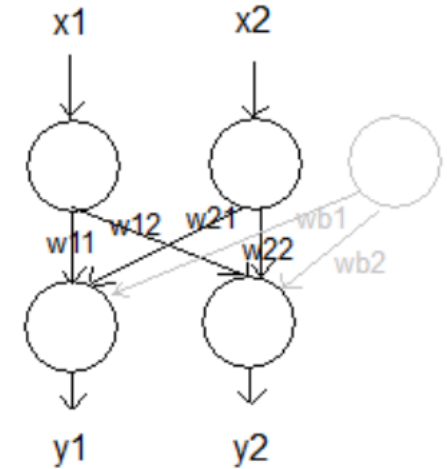
2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.2	0.3	0	0.1	0.2	0.7			0	0
0	1									1	0
1	0									1	0
1	1									1	1

Second epoch

2. SLP with multiple outputs

x1	x2	w11	w21	wb1	w12	w22	wb2	y1	y2	t1	t2
0	0	0.2	0.3	0	0.1	0.2	0.7	0	0	0	0
0	1	0.2	0.3	0	0.1	0.2	0.7	1	0	1	0
1	0	0.2	0.3	0	0.1	0.2	0.7	1	0	1	0
1	1	0.2	0.3	0	0.1	0.2	0.7	1	1	1	1
Next weights		0.2	0.3	0	0.1	0.2	0.7				



Second epoch